QC STORY

SMALL STEPS, BIG IMPACT • AI Demand Forecasting & Dynamic Inventory Optimization

"Predictive Precision Today, Resilient Supply Chain Tomorrow"
ISB AMPBA Capstone Project with Deloitte | Commercial Air Conditioning

Team Name : ISB Capstone Team (Sai Prashanth, Arvind, Hariharan, Vani, Vedika)
Sponsor : Deloitte | Department: Supply Chain & Analytics
Evaluation : FY24-25 Holdout Validated (444K units) | 14/14 Audits Passed

1 THEME / PROBLEM STATEMENT

Title : AI SKU Demand Forecasting & Safety Stock Sizing
Process/Line : Commercial Room AC (RA-ODU) Supply Chain — 5 South Hubs
Problem : 97.1% volatile/lumpy demand & acute Q4 summer surge cause severe stockouts and massive post-season dead-stock under legacy 8-wk moving average.
Period : Jul 2021 – Jun 2024 (Training) to FY25 (Holdout Validation)

2 CURRENT SITUATION (AS-IS)

- What is happening? Acute demand lumpiness (45% annual sales in 10 summer weeks); legacy MA ignores seasonality.
- Where is it occurring? 5 South Hubs: Chennai (MAA 36.7%), Hyderabad (SBD/SBD1 43.4%), Cochin (COK 10%), Bangalore (BLR 9.9%).
- Scale: 99,395 real billing transactions across 240 active ODU SKUs and 1,038 billing days analyzed.

Demand Pattern	**97.1% Volatile (62.3% Erratic, 34.8% Lumpy by Syntetos-Boylan)**
Legacy Forecast	Static 8-week Moving Average (~50% weekly WAPE error)
Working Capital	Excess safety stocks tying up ~20% unneeded buffer capital
Stockout Risk	Recurrent peak stockouts on fast-moving 1.5T/2.0T high-margin SKUs

Demand Spike (As-Is)
Peak: Q4 Summer Surge
- 45% annual vol in 10 wks
- Intermittent weekly zero-sales
- MA causes lag & stockouts

3 TARGET (GOAL)

Target Statement:
Deploy multi-model AI forecasting engine to minimize forecast error and optimize safety stock buffers.
Reduce : WAPE Forecast Error & Excess Buffer
From : Legacy MA (~50-60% weekly error)
To : Monthly WAPE <=34.4% (Acc >=65.6%), Qtr Acc >=71.8%
By Date : FY25 Holdout Deployment

GOAL
Leaner Inventory
Zero Stockouts
Happier Customers

✓ 71.8% Regional Acc
✓ ~20% Buffer Cut
✓ >=95% OTIF Fill

4 ROOT CAUSE ANALYSIS (WHY-WHY / FISHBONE)

Fishbone Analysis (5M + 1E):
- **Method:** Static 8-wk MA; ignores seasonality & demand lumpiness
- **Machine:** Single-series models cannot cross-learn across SKUs
- **Material:** 240 SKUs with erratic velocity & high zero-demand weeks
- **Environment:** Acute summer heatwaves drive 45% annual sales in 10 wks
- **Measurement:** Evaluated on flat averages without holdout rigor or P10/P90

Potential Cause	Verification	Root Cause
Static Moving Average	52-wk holdout backtest	YES (Y)
No Demand Profiling	Syntetos-Boylan test	YES (Y)
Isolated Single Series	LightGBM cross-learning	YES (Y)

Root Cause Identified: Isolated single-series statistical rules cannot learn cross-SKU dynamics or adapt to extreme lumpy intermittency without machine learning and demand quadrant segmentation.

5 COUNTERMEASURE

Root Cause	Action Planned / Countermeasure	Responsibility	Due / Status
Lumpy Intermittency	Syntetos-Boylan demand quadrant classification (Smooth, Erratic, Lumpy, Intermittent)	Data Science	Completed
Single-Series Limits	17-Model Tournament: Global LightGBM cross-learning, SARIMA, ETS, Croston/TSB/SBA	ML Team	Completed
Overfitting Risk	Strict 52-week rolling-origin holdout backtesting; zero lookahead data leakage	Validation Lead	14/14 Passed
Planner Operational Gap	7-Page Interactive Streamlit Cockpit with live forecasting, P10/P90 & report exports	Product Lead	Live Deployed

CHECKLIST
☑ **Eliminate**
Blind factory push & static Excel models
☑ **Prevent**
Peak summer stockouts & post-season dead stock
☑ **Sustain**
Automated drift audit & continuous retraining

6 IMPLEMENTATION

Actions Completed:
1. **Data Pipeline & Cleaning:** Cleaned 99,395 transactions, isolated synthetic sheets, audited 14/14 zero-leakage validation checks.
2. **Multi-Model Architecture:** Built global LightGBM cross-learning model pooling 273 series with lag, rolling & seasonal index features.
3. **Interactive Streamlit Cockpit:** Shipped 7-module dashboard: Executive Cockpit, Forecast Visualizer, Inventory Optimizer, Report Generators.

BEFORE (As-Is)
- Static 8-wk Moving Avg
- Sub-50% weekly accuracy
- Reactive factory push
- Frequent peak stockouts
- ~20% excess safety stock

AFTER (AI Cockpit)
- Global LightGBM cross-learn
- 71.8% Regional Qtr Acc
- 65.6% Monthly Acc
- ~20% leaner safety stock
- Live 7-page interactive UI

7 RESULTS & BUSINESS IMPACT

Parameter / Metric	Before (Legacy MA)	After (AI Engine)	Improvement / Impact
Regional Monthly WAPE	50.5% (MA4 baseline)	34.4% (Rolling LightGBM)	+31.9% Error Reduction
Regional Quarterly Accuracy	55.0% - 60.0%	71.8% Accuracy	+16.8% Higher Precision
Model Forecast Bias	-22.6% (Severe Under-forecast)	+0.6% (Near-Zero Bias)	Eliminates Peak Shortages
Safety Stock Buffer	High excess holding	~20% Excess Buffer Cut	Massive Working Capital Freed

IMPACT
71.8%
Regional Qtr Accuracy
~20% Buffer Reduction
>=95% OTIF Fill Rate

8 STANDARDIZATION, FUTURE SCOPE & HORIZONTAL DEPLOYMENT

Standardization & Governance:
SOP updated: ☑ Yes ☐ No
Training completed: ☑ Yes ☐ No
Control plan updated: ☑ Yes ☐ No

Horizontal Scaling Roadmap:
- Expand model from South Hubs to North, West, and East depots.
- Hierarchical MinT reconciliation for coherent SKU-to-Factory roll-up.

FUTURE SCOPE 1: Dynamic Stock Transfer-In & Transfer-Out Optimization
- **Inter-Branch Network Balancing:** Automated MILP optimization calculating optimal lateral Transfer-In (deficit depots) and Transfer-Out (surplus depots) across the 5 South distribution hubs.
- **Zero-Lead-Time Stockout Mitigation:** When local heatwaves surge demand in one branch (e.g. Chennai MAA), lateral transfer from a neighboring hub (e.g. Bangalore BLR) resolves shortages in 24–48 hours instead of 4–6 week factory lead times.
- **Freight Cost vs. Margin Balancing:** Mathematical algorithm balances secondary transfer freight and transit times against lost margin penalties, reducing total regional network safety stock by 12–15% through virtual pooling.

FUTURE SCOPE 2: Local News & Real-Time AI Market Intelligence Integration
- **GenAI External Signal Crawlers:** Autonomous LLM pipelines ingesting regional news headlines, IMD meteorological heatwave warnings, local commercial real-estate starts, and dealer conference feedback.
- **NLP Demand Shock Multipliers:** Translates unstructured regional news and municipal policy shifts into quantitative demand elasticity multipliers—converting forecasting from backward-looking to proactive leading sensing.
- **Conversational Supply Chain Copilot:** In-cockpit GenAI agent providing causal explanations for demand surges (e.g. 'Chennai demand surged +22% due to announced tech corridor project and heatwave alert').